

COLUMBUS, NE, USA

WMO: 725565

Lat: 41.450N

Lon: 97.345W

Elev: 1447

StdP: 13.94

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94951

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	0.1	-12.8	2.9	-3.8	-8.3	3.7	1.1	34.2	22.3	30.0	21.7	7.6	330	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	93.2	74.8	90.8	74.0	88.3	73.1	78.2	89.1	76.6	87.2	75.0	85.2	12.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.9	138.3	85.4	73.0	129.1	83.6	72.0	124.7	82.5	42.7	89.8	41.1	86.4	39.5	86.0	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	24.2	20.6	-13.1	97.7	4.0	2.7	-16.0	99.7	-18.3	101.3	-20.6	102.8	-23.5	104.8
			-13.8	81.3	3.9	1.4	-16.6	82.3	-18.8	83.1	-21.0	83.9	-23.8	84.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	23.0	26.7	38.4	49.6	60.6	70.9	75.8	72.6	65.5	51.7	37.8	25.8	(6)
(7)		DBStd	20.88	12.21	12.31	11.93	9.87	8.96	6.99	5.78	5.71	8.72	9.60	10.43	11.02	(7)
(8)		HDD50	3249	837	652	390	124	14	0	0	4	94	384	750		(8)
(9)		HDD65	6451	1302	1071	825	469	195	25	3	7	98	425	816	1215	(9)
(10)		CDD50	3245	0	1	31	112	343	627	799	699	470	145	18	0	(10)
(11)		CDD65	975	0	0	1	7	59	203	337	242	114	12	0	0	(11)
(12)		CDH74	10267	0	0	24	139	710	2134	3570	2260	1252	172	6	0	(12)
(13)		CDH80	3979	0	0	4	36	236	811	1545	848	450	49	0	0	(13)
(14)	Wind	WSAvg	9.2	9.4	9.9	10.8	11.6	10.2	8.7	7.4	6.8	8.0	9.3	9.3	9.2	(14)
(15)	Precipitation	PrecAvg	28.3	0.7	0.7	1.6	2.8	4.3	4.8	3.5	3.1	2.4	2.1	1.3	0.9	(15)
(16)		PrecMax	37.9	1.4	1.6	3.7	4.6	8.5	11.2	9.1	9.6	5.9	6.4	3.6	2.8	(16)
(17)		PrecMin	19.8	0.2	0.1	0.0	0.4	1.3	1.6	0.9	0.7	0.5	0.2	0.0	0.0	(17)
(18)		PrecStd	4.8	0.3	0.4	1.0	1.1	2.0	2.4	2.1	2.0	1.5	1.6	1.0	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	64.4	77.2	84.3	91.3	95.2	98.7	96.6	92.8	86.2	72.9	57.0	(19)
(20)			MCWB	44.5	49.4	56.3	62.6	68.2	73.8	76.3	74.8	71.5	66.5	56.2	47.5	(20)
(21)		2%	DB	50.0	55.1	70.4	78.6	85.9	91.0	93.4	91.2	89.8	78.9	65.9	51.7	(21)
(22)			MCWB	41.0	45.1	55.2	59.6	67.0	73.3	76.3	75.6	71.0	62.1	52.8	43.6	(22)
(23)		5%	DB	44.7	49.6	63.4	72.4	81.7	88.2	90.9	88.3	84.9	73.0	60.8	45.9	(23)
(24)			MCWB	37.7	41.8	51.7	57.0	64.9	72.0	75.8	74.7	69.6	58.6	49.5	39.0	(24)
(25)		10%	DB	39.0	44.7	57.2	66.4	77.3	84.2	88.4	84.2	81.4	69.0	54.8	41.2	(25)
(26)			MCWB	34.5	38.4	47.5	54.0	62.9	70.0	74.8	72.9	66.6	57.5	45.9	35.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	52.0	61.9	65.1	72.7	78.2	80.8	80.1	75.5	69.1	58.7	49.9	(27)
(28)			MCD B	57.4	61.3	72.6	79.1	85.0	89.6	91.9	89.9	87.5	81.5	67.8	56.1	(28)
(29)		2%	WB	41.5	46.2	57.2	61.4	69.5	75.6	78.8	77.5	73.2	65.2	54.5	43.2	(29)
(30)			MCD B	49.4	54.6	66.2	75.2	82.6	86.7	90.1	87.6	84.7	74.2	64.5	50.8	(30)
(31)		5%	WB	38.0	41.9	52.8	58.5	67.0	73.8	77.3	75.7	71.1	61.3	50.4	39.4	(31)
(32)			MCD B	43.7	49.3	63.8	70.3	78.2	84.8	88.4	85.4	81.9	70.2	59.6	45.5	(32)
(33)		10%	WB	34.8	38.2	48.3	55.2	64.2	71.6	75.6	74.0	69.0	57.9	46.2	36.1	(33)
(34)			MCD B	39.2	43.9	57.6	66.1	75.2	82.1	86.2	82.9	79.6	67.0	54.5	41.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.4	19.9	22.1	23.2	22.2	21.6	20.8	20.9	24.6	24.0	22.3	19.2	(35)
(36)			MCD BR	27.5	28.5	32.9	31.7	28.6	25.1	23.2	24.5	27.2	31.7	32.7	26.9	(36)
(37)			MCWBR	20.1	19.7	20.4	17.9	14.3	12.3	11.7	12.6	13.2	17.6	20.8	19.9	(37)
(38)		5% WB	MCD BR	26.1	27.2	29.9	27.7	24.8	21.5	20.6	21.2	23.4	25.1	30.4	25.9	(38)
(39)			MCWBR	19.5	19.4	19.2	17.1	14.0	12.2	11.8	12.2	12.6	15.7	20.5	19.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.292	0.320	0.373	0.393	0.406	0.413	0.411	0.373	0.327	0.288	0.278	(40)	
(41)		taud	2.460	2.457	2.415	2.281	2.302	2.323	2.309	2.309	2.370	2.486	2.510	2.470	(41)	
(42)		Ebn at Noon	280	293	295	284	280	276	272	270	273	276	275	271	(42)	
(43)		Edh at Noon	24	28	33	41	41	40	40	39	34	27	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	666	952	1246	1594	1859	2111	2097	1801	1467	1041	735	550	(44)	
(45)		RadStd	56	61	101	95	124	143	116	122	91	110	50	56	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page